Venkata Lakshmi Parimala Pasupuleti

Senior FDE · Frontier GenAI · Agentic Platforms · Client Delivery · LLMOps · HITL · Anthropic · OpenAI · Enterprise AI

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SUMMARY

Senior engineer with 9+ years of production depth and a track record of embedding with clients, translating business problems into Frontier GenAI solutions, and delivering them to production. At Magizhs (forward deployed), embedded with the client, mapped their real workflow, prototyped, and shipped a production agentic platform - from POC to live system with guardrails and audit logging. Built **Leasel**: enterprise AI platform with agentic workflows, tool use, human-in-the-loop controls, RAG pipelines, and responsible AI compliance - the kind of working software FDE pods deliver. Built **AgentProbe**: open-source LLMOps toolkit with evaluation frameworks, model monitoring, prompt management, and CI regression harness - the operational layer that keeps GenAI solutions reliable after they ship. Deep hands-on experience with Anthropic (Claude API, tool use), OpenAI, Azure OpenAI, Gemini, and Vertex AI. At Entrata (9+ years, 100K+ users), delivered enterprise solutions end-to-end across a regulated multi-tenant platform, collaborating directly with senior stakeholders, product owners, and business teams to scope and deliver reusable components. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Forward Deployed Engineering · Client Delivery | Frontier GenAI · Agentic Platforms · Workflows | Anthropic · Claude API · Tool Use

OpenAI · Azure OpenAI · Gemini · Vertex AI | Human-in-the-Loop · HITL Controls · Escalation | LLMOps · MLOps · Evaluation Frameworks

Model Monitoring · Prompt Management | RAG Pipelines · Vector Stores · Embeddings | LangGraph · LangChain · Multi-Agent Orchestration

Enterprise AI Platforms · System Integrations | Prompt Engineering · Structured Outputs · Guardrails | Responsible AI · Safety · Compliance · Model Risk

Python · JavaScript · TypeScript · SQL | AWS · Azure · GCP · Cloud-Native | CI/CD · Testing · Logging · Observability

Stakeholder Engagement · Senior Client Communication | Audit Logging · PHI · PII · Access Control

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, GenAI-powered features, and data pipelines end-to-end; integrating AI components with backend APIs and surfacing business insights into application-facing reporting layers for stakeholders.

AI/ML Engineer Intern (Forward Deployed)

May 2025 - Aug 2025

Magizhs Tech Solution

- Embedded with client to identify business needs, led working sessions to map workflow and shape the solution, then prototyped and delivered a production agentic platform - LangGraph multi-agent workflow with RAG pipeline, FastAPI serving, validation guardrails, compliance boundaries, and audit logging throughout.
- Deployed MCP server connecting agent outputs to enterprise systems; defined API contracts, built failure routing and exception paths; contributed reusable components and runbook documentation for client handoff and long-term maintainability.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, enterprise multi-tenant)

- **Collaborated directly with senior stakeholders, product owners, and architects** to scope, prioritize, and deliver enterprise AI-adjacent solutions - translated complex regulated business requirements into production software; built trusted relationships and managed expectations across product, operations, compliance, and business teams.
- **Delivered production-quality code across 20+ enterprise modules** - CI/CD pipelines, automated testing, structured logging, versioning, and documentation; sprint sizing and delivery aligned with senior team members across 120+ Agile sprints; contributed reusable components and reference implementations.
- **Owned regulated data pipeline end-to-end** - HUD compliance ETL (federal rows → 180,000 domain records) powering eligibility and rent calculations for 100K+ housing units; PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage, zero security incidents); zero violations across 7 years.
- Built observability tooling (40% debugging time reduction); owned production incident response and postmortems; Python/Bash automation eliminating 80% manual workload; mentored 5+ engineers; 300+ code reviews.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built enterprise backend modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration; delivered large-scale data import and reconciliation workflows (1.6M records).

PROJECTS

Leasel - Enterprise Agentic AI Platform (Tool Use · HITL · RAG · Responsible AI) 2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Three-surface enterprise AI platform: **agentic workflow** (8-agent LangGraph pipeline with orchestration, tool use, durable state, and conditional routing); **RAG pipeline** (hybrid search, semantic reranking, citation grounding, vector store retrieval); **self-service interface** with hard safety boundaries enforced in code. Architecture decisions balance quality, safety, latency, and model risk - responsible AI throughout: HITL escalation on every regulated decision, audit logging on every agent action, access-controlled retrieval.

- CI/CD regression gate; 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

AgentProbe - LLMOps Evaluation Framework & Model Monitoring Toolkit 2025 I Python, FastAPI, LangGraph, CI/CD I github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLMOps toolkit: evaluation frameworks with golden datasets and LLM-as-judge scoring; model monitoring and post-launch observability; prompt management across providers; CI-integrated regression harness; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); multi-provider routing (Anthropic, OpenAI, Azure OpenAI, Gemini, Groq, Ollama). The operational layer that keeps Frontier GenAI solutions reliable after they ship.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017